Demo Client · DEMO-0000
11 November 2026 · v1.0

| Field | Value |
|---|---|
| Document | Methodology for independent strategy validation |
| Template version | strategy_validation 1.0.0 / proquant-core 0.1.0 |
| Engagement | DEMO-0000 · Demo Client |
| Strategy under review | Dual moving-average trend (long-only) (fast=20,slow=100) |
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| Run ID(s) | sv_20261111T141952Z_46f599 |
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## 1. Purpose and scope

The validation assesses whether the historical performance of the strategy is (a) correctly computed, (b) achievable after realistic costs and execution, (c) robust out of sample and to parameter changes, and (d) unlikely to be the product of overfitting or data biases. It does not assess the investment merit of the strategy, its capacity at a given AUM (unless in scope), operational readiness, or future performance.

**Strategy under review:** Dual moving-average trend (long-only) with parameters fast=20,slow=100; execution lag 1 day(s); costs 5.0 bps per trade.

## 2. Data

### 2.1 Inputs

| Data set | Source | Size | Period | Fingerprint (SHA-256) |
|---|---|---|---|---|
| prices | prices.csv | 99,769 rows | 2012-01-02 to 2025-12-31 | `b774dec28e812fb9` |
| universe | universe.csv | 30 rows | 2012-01-02 to 2025-12-31 | `d44d49659ae7dd4c` |
| benchmark | benchmark.csv | 3,653 rows | 2012-01-02 to 2025-12-31 | `50d69ecafe672230` |

Each input is fingerprinted with SHA-256 on ingest (file bytes and DataFrame content) and recorded in the run audit trail with the source description (path, query, endpoint).

### 2.2 Quality checks

Per input: pandera schema (types, positivity, OHLC consistency, uniqueness), missing values (error above `quality.max_missing_pct`), exact and conflicting duplicate keys, robust outliers on daily returns (|x – median| / (1.4826·MAD) > `quality.outlier_z`). Reports: `data_quality/dq_<dataset>.json|html`.

### 2.3 Cleaning rules (all logged in `tables/cleaning_log.csv`)

1. Rows with unparseable dates or missing identifiers are dropped.
2. Non-positive prices are set to missing.

3. Exact duplicate rows are dropped; for conflicting duplicates of (date, ticker) the last record is kept and each case logged.
4. **Bad ticks:** a daily move larger than `bad_tick_threshold` that reverses the next day (compounded two-day move within half the threshold) is set to missing. Prices are never "corrected".
5. **Stale prices:** runs of identical closes ≥ `stale_days` are flagged, not changed.
6. **Point-in-time universe:** prices before `listed_from` and on/after `delisted_on` are removed.
7. Interior gaps up to `max_ffill_days` are forward-filled; no extrapolation after the last observed price. After cleaning, the long panel is re-checked and must have zero errors.

## 3. Backtest engine

### 3.1 Timing

Let w_t be the target weight decided with information up to the close of day t. With execution lag L, the trade executes at the close of day t+L and the position earns r_{t+L+1}. Hence the position earning on day s is p_s = w_{s−1−L}. L = 0 corresponds to market-on-close execution on the signal day (optimistic); L = 1 (default) to execution at the next close. Returns are simple close-to-close returns of adjusted prices.

### 3.2 Portfolio return

Gross return R_s = Σ_i p_{s,i} · r_{s,i}. A held position whose price is missing earns 0 on that day. Unallocated capital earns the risk-free rate only through the Sharpe calculation (configurable, default 0).

### 3.3 Turnover and costs

Pre-trade weights are the previous positions after drift: d_{s,i} = p_{s−1,i}(1 + r_{s−1,i}) / (1 + R_{s−1}). Turnover T_s = Σ_i |p_{s,i} − d_{s,i}|. Trading cost C_s = T_s · (commission + spread/2 + slippage) / $10^4$ (bps of traded notional). Borrow cost B_s = Σ_i max(−p_{s,i}, 0) · borrow_bps / $10^4$ / 252. Net return N_s = R_s − C_s − B_s. Break-even cost c *(bps per trade) solves* $\Sigma R_s - c \cdot \Sigma T_s - \Sigma B_s = 0$.

## 4. Out-of-sample testing

### 4.1 Walk-forward

Rolling windows: train `train_years`, test `test_years`, stepping by the test length (anchored option available). In each training window the parameter set with the highest net Sharpe ratio is selected from the grid and applied, unchanged, to the following test window. Test windows are concatenated into one out-of-sample (OOS) series. Walk-forward efficiency (WFE) = Sharpe(OOS series) / mean(in-sample Sharpe of selected sets). Because the strategy is causal (verified by §5.1), weights computed on the full history equal those computed on data available at each date, so the in-sample selection only uses past returns.

### 4.2 Hold-out

The period from `holdout_start` was confirmed by the client as not used in research. The chosen parameters are evaluated separately on the development period and the hold-out; the ratio of Sharpe ratios is reported.

# 5. Overfitting and bias diagnostics

## 5.1 Look-ahead bias

*Truncation test:* for k random cut-off dates T, the strategy is recomputed on data up to T; the weights at T must equal those computed on the full data set (max absolute difference ≤ 1e-10). Any difference proves use of future information. *Execution-delay sensitivity:* net Sharpe for lags {0,1,2,3}; a sharp drop with one extra day indicates fragility to implementation timing (or subtle look-ahead in the data, e.g. signals computed on prices published after the close).

## 5.2 Survivorship bias

The backtest uses the point-in-time universe including delisted securities. It is re-run on survivors only (securities never delisted). The CAGR difference estimates the bias a survivor-only data set would have introduced. If the data contains no delisted securities, the bias cannot be assessed and the test is graded WARN.

## 5.3 Data snooping / multiple testing

With per-period Sharpe ratio SR, n observations, skewness $\gamma_3$ and (raw) kurtosis $\gamma_4$:

Probabilistic Sharpe Ratio (Bailey & López de Prado, 2012): $PSR(SR) = \Phi\left( (SR - SR) \cdot \sqrt{(n - 1)} / \sqrt{(1 - \gamma_3 \cdot SR + (\gamma_4 - 1)/4 \cdot SR^2)} \right)$.

Expected maximum Sharpe of N independent zero-skill trials with variance V of trial Sharpe ratios (False Strategy Theorem): $SR_0 = \sqrt{V} \cdot \left( (1 - \gamma) \cdot \Phi^{-1}(1 - 1/N) + \gamma \cdot \Phi^{-1}(1 - 1/(N \cdot e)) \right)$, $\gamma = 0.5772$ (Euler–Mascheroni).

Deflated Sharpe Ratio (Bailey & López de Prado, 2014): $DSR = PSR(SR_0)$. N = grid size + trials declared by the client; V from the dispersion of grid Sharpe ratios (or an assumed annual standard deviation when no grid exists). Minimum Track Record Length: $MinTRL = 1 + (1 - \gamma_3 \cdot SR + (\gamma_4 - 1)/4 \cdot SR^2) \cdot (z_\alpha / (SR - SR^*))^2$.

## 5.4 Probability of Backtest Overfitting (Bailey, Borwein, López de Prado & Zhu, 2017)

The T×N matrix of daily net returns for the N grid variants is split into S contiguous blocks ( `pbo.n_splits` ). For each of the C(S, S/2) combinations, the blocks form an in-sample set and the complement an OOS set. The variant with the best IS Sharpe is located in the OOS ranking: $\omega = \text{rank}/(N+1)$, $\lambda = \ln(\omega/(1-\omega))$. PBO = share of combinations with $\lambda \le 0$, i.e. the probability that the IS-optimal variant ranks below the OOS median.

# 6. Performance and risk metrics

Annualisation with 252 periods per year. CAGR = (Π(1+r))^{252/n} − 1; volatility = σ·√252; Sharpe = mean excess return / σ · √252; Sortino uses downside deviation √mean(min(r,0)²); maximum drawdown from the running peak of the equity curve; drawdown duration = longest run below a previous peak (trading days); Calmar = CAGR / |max drawdown|; hit rate = share of non-zero days with positive return; historical VaR/ES at 95% (one day, positive loss); beta and alpha from OLS against the benchmark; tracking error and information ratio on active returns; annual turnover = mean daily turnover × 252.

# 7. Robustness

- **Parameter sensitivity:** net Sharpe, CAGR, drawdown and turnover over the grid. Stability statistics: share of the grid with positive Sharpe; ratio of the weakest neighbouring setting to the chosen one (plateau vs peak); rank of the chosen setting.
- **Cost stress:** costs scaled by `cost_multipliers` (0×–3×).

- **Monte Carlo:** stationary bootstrap (Politis & Romano, 1994) of daily net returns with geometric block lengths (mean `mean_block_days`), `n_sims` paths of the original length, fixed seed. Reported: percentiles of CAGR, Sharpe and maximum drawdown; probability of a negative Sharpe.

## 8. Scorecard and overall assessment

Each test is graded against thresholds agreed at kickoff (defaults in `scorecard.py`): PASS at or beyond the pass threshold, WARN between warn and pass, FAIL beyond warn, N/A when not applicable. Overall: any FAIL → "Not validated"; otherwise any WARN → "Validated with reservations"; otherwise "Validated". The assessment is an opinion on the robustness of the evidence, not on future returns.

## 9. Assumptions, limitations and deviations

Standard limitations: linear cost model (no size-dependent market impact; capacity not established); daily close simulation (no intraday fills, halts, partial fills); delisting returns captured only to the last traded price; DSR depends on disclosed trials; bootstrap cannot generate regimes absent from history; thresholds are conventions. No deviations from the standard methodology.

## 10. Reproducibility

Each run folder contains `config_used.yaml`, `audit.json` (analyst, timestamps, git commit of the templates repository, package versions, input fingerprints, output hashes), `run.log`, every result table as CSV and the report. Re-running the pipeline with the same inputs, configuration and code version reproduces all results (Monte Carlo uses a fixed seed).

## 11. References

- Bailey, D. H., & López de Prado, M. (2012). The Sharpe Ratio Efficient Frontier. *Journal of Risk*, 15(2).
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- Bailey, D. H., Borwein, J., López de Prado, M., & Zhu, Q. J. (2017). The Probability of Backtest Overfitting. *Journal of Computational Finance*, 20(4).
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- Harvey, C. R., & Liu, Y. (2015). Backtesting. *Journal of Portfolio Management*, 42(1).
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